NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-20/36E
6 May 1988
SUPERSEDING
MIL-C-20/36D
19 September 1983

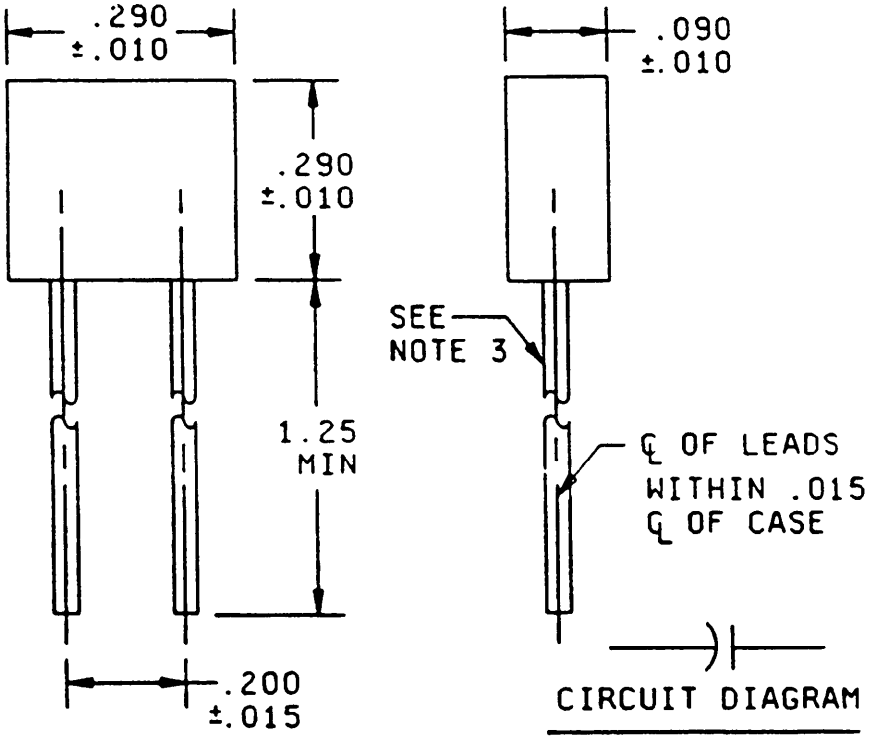
PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR06 AND CC06.

Style CC06 is inactive for new design after 19 September 1983. Use style CCR06.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-20.



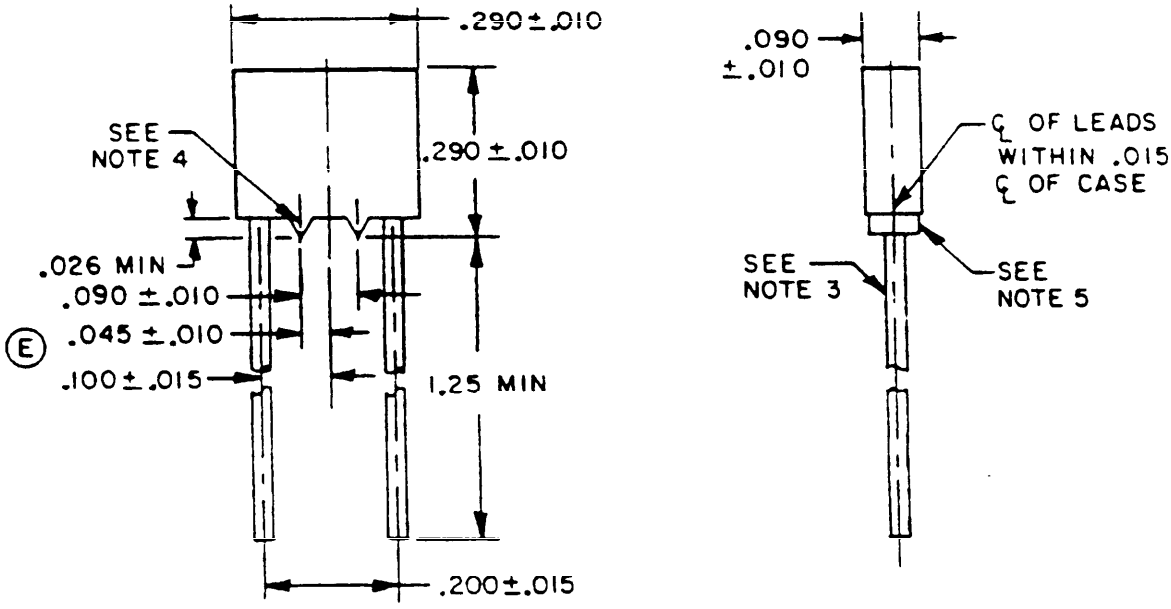
CONFIGURATION WITHOUT STANDOFFS

FIGURE 1. Styles CCR06 and CC06 capacitors.

(E) denotes changes

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OPTIONAL CONFIGURATION WITH STANDOFFS (CCR06 ONLY)

Inches	mm
.010	0.25
.015	0.38
.026	0.66
.045	1.14
.090	2.29
.100	2.54
.200	5.08
.290	7.37
1.25	31.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
Ⓔ 3. Lead diameter shall be .023 (0.58 mm) to .029 (0.74 mm).
Ⓔ 4. Optional standoffs for style CCR06 shall be of triangular shape so that the standoffs will provide line contact with the surface upon which the capacitor is mounted. The shape of the triangle is optional.
Ⓔ 5. Thickness of the standoffs shall be .075 (1.90 mm) to the body thickness.
Ⓒ 6. Lead length may be a minimum of .625 (15.88 mm) for use in tape and reel automatic insertion equipment, when specified.
Ⓔ 7. At the option of the user, the standoff configuration may be furnished as a replacement for the nonstandoff configuration of the same style.

FIGURE 1. Styles CCR06 and CC06 capacitors - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Lead type: Radial.

Case type: Molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: CG.

Failure rate level (CCR06 only): M (1.0%), P (0.1%), R (.01%), or S (.001%).

Thermal shock and voltage conditioning (CCR06 only): In accordance with MIL-C-20.

Capacitance: Within tolerance specified (see table I).

Ⓔ TABLE I. Capacitor characteristics.

Type designation 1/	Rated voltage	Nom. cap.	Cap. tol.	Type designation 1/	Rated voltage	Nom. cap.	Cap. tol.
	volts, dc	pF			volts, dc	pF	
CC-06CG361---	200	360	FGJ	CC-06CG202---	100	2000	FGJ
CC-06CG391---	"	390	"	CC-06CG222---	"	2200	"
CC-06CG431---	"	430	"	CC-06CG242---	"	2400	"
CC-06CG471---	"	470	"	CC-06CG272---	"	2700	"
CC-06CG511---	"	510	"	CC-06CG302---	"	3000	"
CC-06CG561---	"	560	"	CC-06CG332---	"	3300	"
CC-06CG621---	"	620	"	CC-06CG362---	"	3600	"
CC-06CG681---	"	680	"	CC-06CG392---	"	3900	"
CC-06CG751---	"	750	"	CC-06CG432---	"	4300	"
CC-06CG821---	"	820	"	CC-06CG472---	"	4700	"
CC-06CG911---	"	910	"	CC-06CG512---	50	5100	FGJK
CC-06CG102---	"	1000	"	CC-06CG562---	"	5600	"
CC-06CG112---	"	1100	"	CC-06CG622---	"	6200	"
CC-06CG122---	"	1200	"	CC-06CG682---	"	6800	"
CC-06CG132---	"	1300	"	CC-06CG752---	"	7500	"
CC-06CG152---	"	1500	"	CC-06CG822---	"	8200	"
CC-06CG162---	"	1600	"	CC-06CG912---	"	9100	"
CC-06CG182---	"	1800	"	CC-06CG103---	"	10000	"
				CC-06CG123---	"	12000	"
				CC-06CG153---	"	15000	"
				CC-06CG183---	"	18000	"

1/ Complete type designation will include the following:

1st dash: Symbol "R" (for style CCR06) or dash will be deleted (for style CC06).
2nd dash: Applicable capacitance tolerance symbol.
3rd dash: Applicable failure rate level symbol (CCR06 only) or dash will be deleted (for style CC06).
Ⓔ 4th dash: Symbol "V" for standoffs (CCR06 only) or dash will be deleted (for style CCR06 without standoffs and style CC06).

Dissipation factor: In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.

Body insulation: Test II.

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...e laws and regulations unless a specific exemption has been obtained. ...applicab

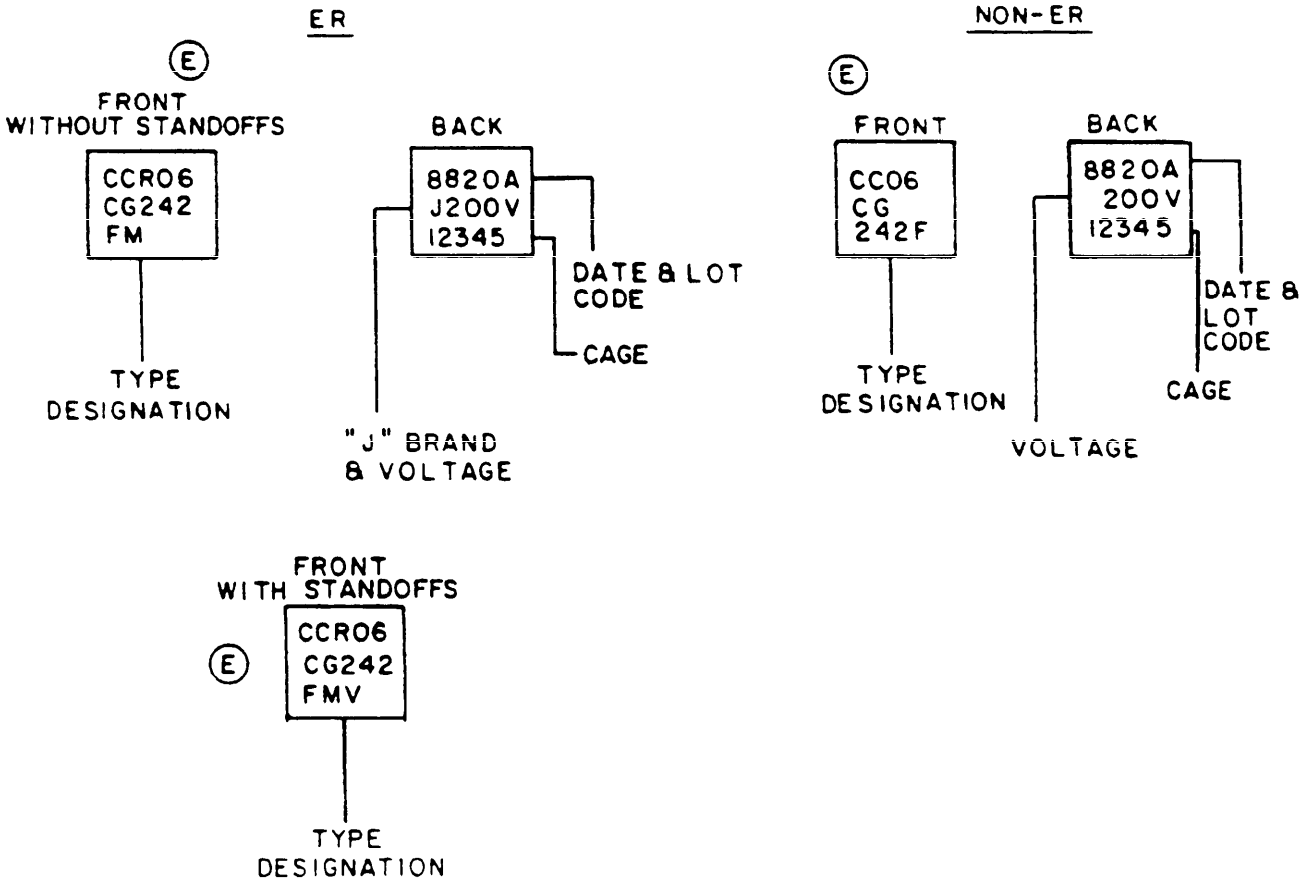
(E) Solderability: For style CCR06 with standoffs, the leads shall be solderable to within .024 (0.61 mm) from the body egress. For style CCR06 without standoffs and style CC06, solderability shall be in accordance with MIL-C-20.

(E) Resistance to soldering heat: In accordance with method 210 of MIL-STD-202, test condition D.

(E) Workmanship: For style CCR06 with standoffs, exposed copper or bare leads shall be permitted to a maximum of .020 (0.51 mm) from the body egress. For style CCR06 without standoffs and style CC06, workmanship shall be in accordance with MIL-C-20.

Life: In accordance with MIL-C-20, operating condition 2.

(E) Marking: Method I of MIL-STD-1285. At the option of the manufacturer, the marking may be placed on one side of the capacitor, or the marking may be placed on two lines as long as it is the same order as shown in the examples.



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Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - OS
Air Force - 99
DLA - ES

User activities:
Navy - AS, MC, SH
Air Force - 11

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5910-1587)

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